

Internship Report

AWS in Cybersecurity

Introduction

Today's internship session was about Amazon Web Services (AWS). Instead of learning only the theoretical concepts of AWS services, I explored how AWS can be applied in the field of cybersecurity. Organizations must protect sensitive data, applications, and cloud resources from cyber threats. AWS provides secure and scalable cloud services that help build strong cybersecurity solutions.

Why Cloud Computing?

Managing security infrastructure on physical servers requires significant investment in hardware, maintenance, updates, and monitoring. AWS allows organizations to use cloud-based security services, scale resources as needed, and pay only for what they use. This improves flexibility while reducing operational costs.

Amazon EC2

Amazon EC2 can host cybersecurity applications such as SIEM platforms, intrusion detection systems, vulnerability scanners, and security monitoring tools. Security teams can quickly increase computing resources during large-scale security scans or incident investigations.

AWS Lambda

AWS Lambda automates security tasks. It can send alerts when suspicious activities are detected, isolate compromised resources, trigger incident response workflows, and perform automated security checks without managing servers.

Amazon S3

Amazon S3 securely stores security logs, audit reports, backup files, malware samples, and digital evidence. High durability and encryption features help protect sensitive security data while ensuring availability.

Database Services

Amazon RDS can store user accounts, incident records, compliance reports, and vulnerability data. DynamoDB can be used where fast access is required for threat intelligence or security events. Managed databases reduce maintenance and improve reliability.

IAM and Security

AWS Identity and Access Management (IAM) controls who can access cloud resources. Administrators assign permissions based on roles, ensuring users receive only the access they need. This reduces unauthorized access and strengthens overall cloud security.

Monitoring and Protection

AWS CloudTrail records API activity for auditing, while Amazon CloudWatch continuously monitors system performance and security events. AWS Shield protects against DDoS attacks, and AWS WAF helps block common web attacks, improving application security.

Conclusion

From today's internship session, I understood that AWS plays a vital role in modern cybersecurity. Services such as EC2, Lambda, S3, RDS, IAM, CloudTrail, CloudWatch, Shield, and WAF work together to build secure, scalable, and reliable cloud environments. This practical example helped me understand how AWS services solve real-world cybersecurity challenges.